Sierra Leone: Fiscal Sustainability Analysis

A. Main results

This appendix summarizes the results of the fiscal sustainability analysis for Sierra Leone.[43] The baseline scenario assumes a sustained high output growth together with low inflation. Real output is projected to grow between 6-7 percent annually in the medium term, with inflation reverting to low single digits.

The primary balance is projected to strengthen over the medium-term. This is brought about, on the one hand, by stronger domestic revenue effort and augmentation of the revenue base. An improved revenue effort will be necessary because the share of foreign grants as percent of GDP is projected to decline from 40 percent of total revenues and grants in 2003 to below 10 percent by year 2022. Consequently, current expenditures would need to be covered to large extent by domestic revenues.

On the other hand, government expenditures as a whole are projected to decline to facilitate the attainment of fiscal sustainability in the long run. The reduction should be achieved in part through cost cutting, rationalization, and other reforms to contain costs and improve efficiency of government services. Furthermore, in light of the need to limit expenditures, outlays need to be prioritized reflecting developmental and poverty-related needs (in the context of the PRSP/MDGs, which provide key guidance on this). The share of investment expenditure is also projected to increase to promote high output growth.

The long-run debt-to-GDP and debt-to-revenue ratios are projected to fall in the medium- to long-run (Table 1). As a result of fiscal consolidation and improved fiscal management, staff projects that debt-to-GDP and debt-to-revenue ratios would remain broadly stable or decline in the long run. In part, this is assisted by debt forgiveness under the HIPC Initiative, which will lower the net present value of external public debt. Moreover, lower domestic borrowing (as percent of GDP) helps to alleviate the rising cost of borrowing from domestic sources, a key medium-term risk to fiscal sustainability in Sierra Leone (the share of domestic debt to total debt stock is projected to fall from 41 percent in 2003 to 18 percent in 2022).

B. Sensitivity Analysis

The overall macroeconomic environment in which the government is operating will have an impact on the fiscal outcome, and by extension fiscal sustainability. That is, changes in real growth and inflation, as well as in the external environment, would need to be offset by fiscal adjustments to preserve long-term fiscal sustainability. Private sector consumption and saving decisions affect interest rates and the cost and level of private investment activity, and eventually economic growth and government's revenue collections and ability to spend on

[43] More details are provided in Chapter 2 of the Selected Issues and Statistical Appendix, 10/29/04 (www.imf.org).

various public programs. Foreign investments in the country would be equally important for supporting domestic activities and export growth. Expectations and public confidence in the government's ability to implement appropriate economic policies influence private sector decisions and hence need to be fostered. Improved macroeconomic performance and fiscal balance would be essential for medium-term fiscal sustainability. The following conclusion can be made regarding the sensitivity analysis:

- Based on historical averages for real growth and primary fiscal balance, the debt indicators would deteriorate in the long run. Although this reflects the country's volatile past, it also suggests that Sierra Leone needs to maintain a high level of output growth and achieve broadly balanced primary position in the medium-term; otherwise, the fiscal position may not be sustainable. The outcome that further fiscal consolidation will be required is reinforced by the results of a scenario where the fiscal primary balance is kept constant at its level in 2002 leads to an explosive trend.

- A temporary negative output shock could lead to an unsustainable fiscal adjustment and rapidly rising debt-to-GDP and debt service ratios. This is because total expenditures, as a percent of GDP, would rise significantly, leading to a larger primary deficit and borrowing.

- A temporary worsening of the primary balance, which could be caused, for example, by delays in external grant financing, may not necessarily affect the government's long-term fiscal position permanently, provided that the authorities adequately adjust the fiscal portfolio in the medium-term.

Furthermore, a liquidity risk could arise from a higher rollover of short-term domestic debt. In Sierra Leone, virtually all domestic public debt is issued as short-term.[44] Since there are no long-term debt instruments available in Sierra Leone (beyond the maturity of one year), all future borrowing would be financed through the issuance of short-term debt. In the long-run, this would result in a situation where virtually all domestic debt would be short-term. Given the size of the domestic debt stock at end-2003, equal to about 41 percent of GDP, the debt carries a significant rollover risk.[45] The government should therefore promote domestic debt markets.

Privatization of state-owned enterprises could also promote fiscal sustainability. The sale of state assets could be used to raise additional revenues to finance poverty-related expenditures or lower the domestic debt stock.

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[44] The remainder comprises noninterest-bearing debt in the Bank of Sierra Leone balance sheet.

[45] The instrument used to refinance domestic borrowing is 3-month Treasury bills, the debt stock would be rolled over four times in a year.

Table 1. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2001-2022
(In percent of GDP, unless otherwise indicated)

	Actual				Estimate	Projections								
	2001	2002	Historical Average 5/	Standard Deviation 5/	2003	2004	2005	2006	2007	2008	2003-08 Average	2013	2022	2009-22 Average
Public sector debt 1/	79.7	72.3	180.4	73.7	76.3	73.9	77.3	78.0	76.3	76.9	76.5	76.8	65.9	73.7
o/w foreign-currency denominated	29.5	27.5	142.8	68.9	35.5	39.2	45.8	49.7	51.8	54.8	46.1	57.7	50.5	55.7
Change in public sector debt	-192.5	-7.4	-11.2	71.1	4.0	-2.4	3.4	0.7	-1.7	0.6	0.8	-0.8	-3.3	-0.8
Identified debt-creating flows	7.5	-13.9	-23.3	59.4	-2.7	-2.0	-5.1	-4.6	-5.4	-4.7	-4.1	-2.8	-4.5	-3.8
Primary deficit	6.0	2.9	3.9	1.6	1.7	2.1	0.5	0.4	-0.8	-0.1	0.6	1.2	-0.9	0.1
Revenue and grants	18.9	20.4	14.3	4.6	20.1	21.3	21.0	22.0	23.4	23.0	21.8	20.8	19.6	20.6
of which : grants	5.9	8.2	3.2	2.6	7.7	9.1	8.1	7.5	7.8	6.6	7.8	3.5	1.6	3.1
Primary (noninterest) expenditure	25.0	23.3	18.2	4.7	21.8	23.4	21.5	22.3	22.6	22.9	22.4	21.9	18.6	20.7
Automatic debt dynamics	1.6	-16.8	-27.1	58.9	-4.4	-4.1	-5.6	-4.9	-4.6	-4.5	-4.7	-4.0	-3.6	-3.8
Contribution from interest rate/growth differential	-43.6	-16.6	-23.2	25.2	-6.8	-5.7	-6.1	-5.7	-5.5	-5.5	-5.9	-5.0	-4.5	-4.9
of which : contribution from average real interest rate	-2.2	0.6	-4.3	3.9	-0.7	-0.5	1.4	0.8	0.3	-0.1	0.2	-0.6	-0.6	-0.6
of which : contribution from real GDP growth	-41.5	-17.2	5.2	27.0	-6.1	-5.2	-7.6	-6.5	-5.8	-5.3	-6.1	-4.4	-3.9	-4.3
Contribution from real exchange rate depreciation	45.3	-0.2	-3.9	43.9	2.4	1.6	0.6	0.8	0.9	0.9	1.2	...	...	1.0
Other identified debt-creating flows	-0.1	0.0	-0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Privatization receipts (negative)	-0.1	0.0	-0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	-200.0	6.5	12.1	88.2	6.7	-0.3	8.4	5.3	3.8	5.2	4.8	2.0	1.2	3.0
NPV of public sector debt	63.5	54.7	106.2	63.6	50.7	44.5	34.4	31.2	27.5	25.1	35.6	21.9	17.8	20.9
o/w foreign-currency denominated	13.3	10.0	68.5	57.4	9.8	9.8	2.9	2.9	3.0	3.0	5.2	2.8	2.4	2.8
o/w external	13.3	10.0	76.1	55.3	9.8	9.8	2.9	2.9	3.0	3.0	5.2	2.8	2.4	2.8
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Gross financing need 2/	33.3	20.4	18.9	8.1	20.5	22.0	18.1	16.8	14.6	13.2	17.5	15.0	12.0	13.7
NPV of public sector debt-to-revenue ratio (in percent) 3/	335.6	268.9	979.1	903.4	252.0	208.6	163.6	142.0	117.5	109.3	165.5	105.5	91.0	101.1
o/w external	70.3	49.1	731.8	661.7	48.9	45.8	13.8	13.3	12.8	13.1	24.6	13.6	12.5	13.5
Debt service-to-revenue ratio (in percent) 3/ 4/	68.6	18.2	39.8	19.9	19.2	24.8	18.4	12.1	11.8	11.0	16.2	19.0	8.3	14.0
Primary deficit that stabilizes the debt-to-GDP ratio	198.5	10.3	74.3	107.6	-2.3	4.5	-2.9	-0.4	0.9	-0.7	-0.1	1.9	2.4	0.9
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	17.9	27.5	-0.9	15.6	9.3	7.4	9.4	7.1	6.5	6.0	7.6	6.0	6.0	6.1
Average real interest rate on domestic currency debt (in percent)	3.7	10.4	-7.3	10.8	-0.9	-3.6	2.3	3.0	3.0	2.9	1.1	2.5	2.5	2.6
Real exchange rate depreciation (in percent, + indicates depreciation)	30.1	6.9	-0.4	24.2	9.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.0	-3.7	23.2	22.3	8.2	14.4	6.9	5.6	4.8	3.8	7.3	3.8	3.8	3.8
Growth of real primary spending (deflated by GDP deflator, in percent)	13.4	10.2	4.3	30.2	2.4	19.9	4.3	13.8	9.0	8.1	9.6	3.1	4.9	4.0
Grant element of new external borrowing (in percent)	56.9	46.0	10.3	21.9	48.0	47.1	42.4	46.5	43.5	40.8	44.7	56.9	56.9	56.9

Sources: Country authorities; and Fund staff estimates and projections.

1/ Public sector comprises the central government or nonfinancial public sector. External debt is reported inclusive of unconditional HIPC debt relief.

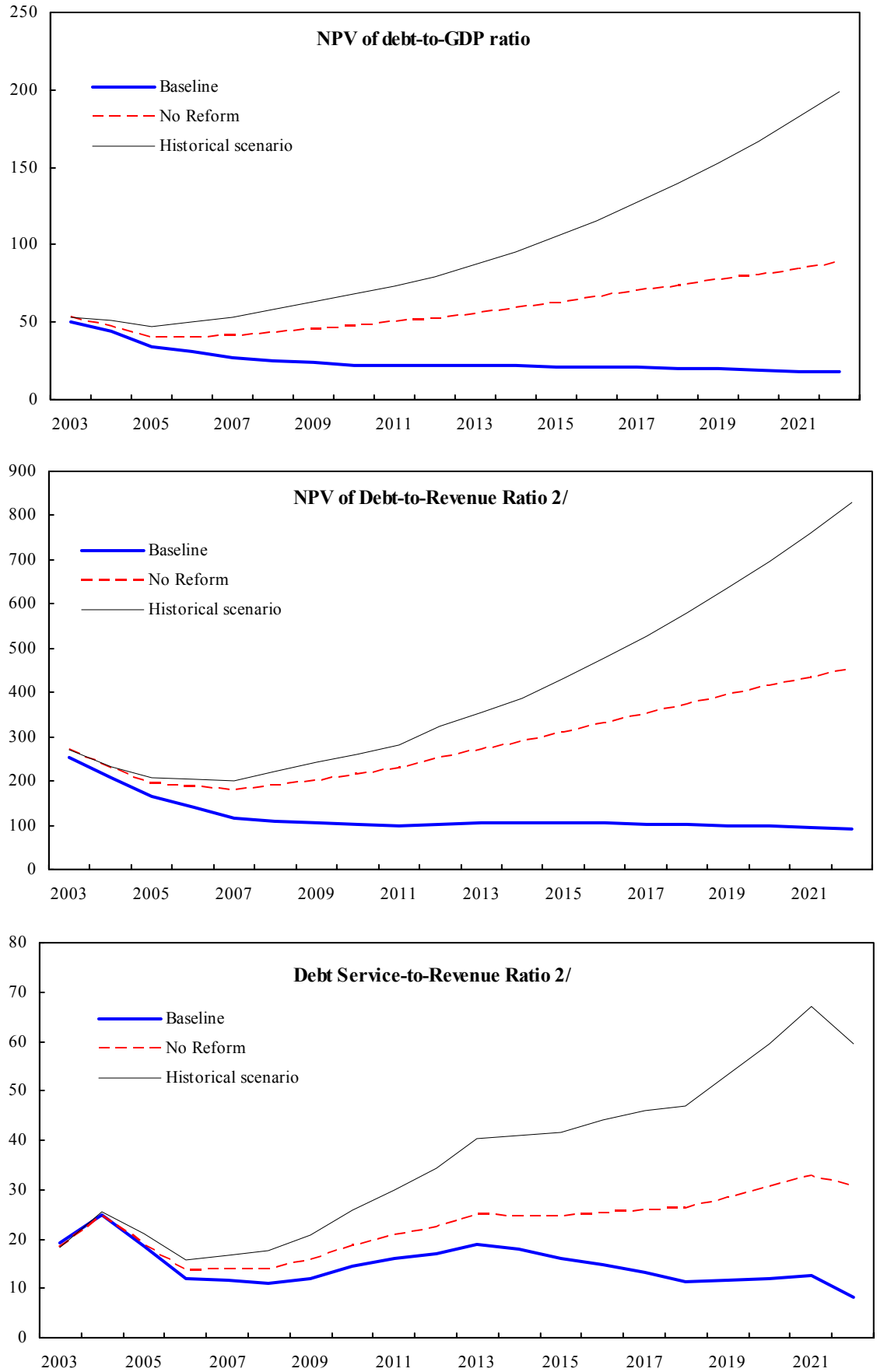
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally bsed on data for period 1993-2002.

Figure 1. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2003–2022 1/

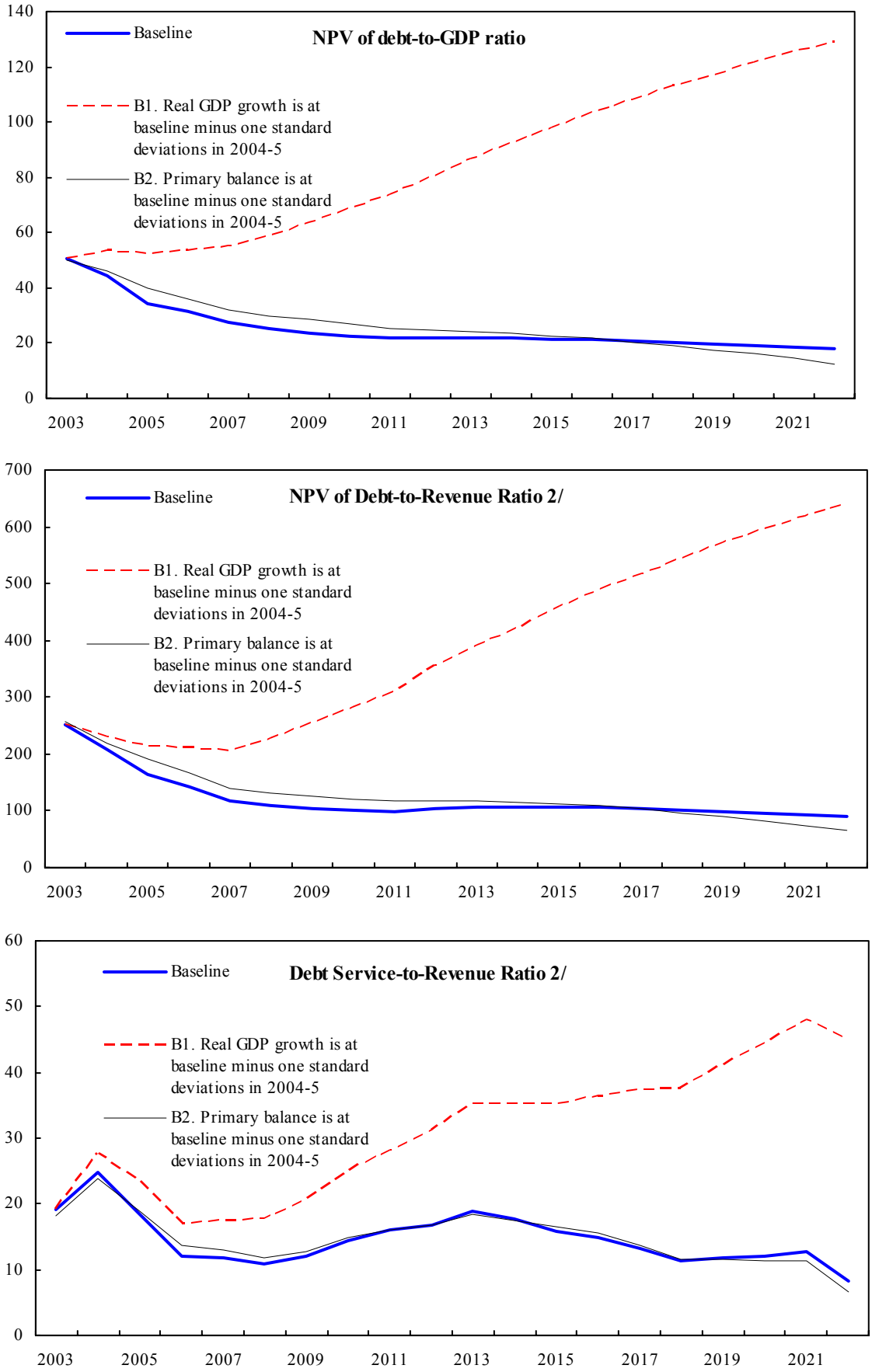


Source: Staff projections and simulations

1/ Historical scenario is based on historic averages for real growth and primary balance over the period 1993-2002

2/ Revenue including grants (excl. domestic debt service)

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2003–2022 1/



Source: Staff projections and simulations.
1/ Historical scenario is based on averages over the period 1993–2002.
2/ Revenue including grants (excl. domestic debt service).

Sierra Leone: External Debt Sustainability Analysis

This section presents the results of updating Sierra Leone's external debt sustainability analysis (DSA). The update of the baseline debt ratios has been done in two steps; first, the decision point DSA has been updated, and then, second, the new DSA framework for low-income countries has been applied (see Box 1).[46] The new framework yields estimates for the NPV of debt-to-exports ratio that are somewhat lower than those obtained from calculations using the HIPC methodology.[47] The baseline scenario under the new framework yields a path for the NPV of debt-to-exports ratio that is fairly stable around 120 percent (Table 1a and Figure 1, middle panel).[48] The profile for the NPV of debt-to-GDP ratio is somewhat different, with the ratio slowing rising to about 34 percent in 2010, and then gradually declining to about 25 percent in 2020 (Figure 1, top panel). The debt service ratio stays for most of the projection period below 10 percent (Figure 1, bottom panel).[49]

The NPV of debt-to-exports and -GDP ratios can be considerate moderate, and the debt service ratio is low. For example, the NPV of debt-to-exports and the NPV of debt-to-GDP ratios in non-HIPC low-income countries in 2001 have been 143 and 39 percent respectively, which are both higher than the projected ratios for Sierra Leone after delivery of HIPC relief.[50] Sierra Leone's debt service ratio of over the medium-term is projected to average about 7 percent, which is only about half the prevailing ratio in non-HIPC low-income countries.[51] The baseline scenario alone seems to suggest that Sierra Leone is at low risk of debt distress.

The application of stress tests reveals, however, substantial vulnerability to shocks. The stress tests comprise an alternative scenario, where key variables grow in the projection period from 2002 onwards at their historic rates, and shocks to individual variables.[52] The

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[46] All debt ratios assume full and unconditional delivery of HIPC relief, and additional bilateral relief beyond HIPC, beginning in 2000.

[47] As discussed in Box 1, this is mainly because of different discount rate assumptions, and because the new framework uses current exports instead of the backward-looking three year average used under HIPC.

[48] Under the HIPC methodology, the NPV of debt-to-exports ratio stays around 145 percent.

[49] The increase in the debt service ratio in the period 2010-15 is due to projected large amortization payments falling due to the Fund in this period, and a slowdown in export growth as kimberlite and rutile mines reach the end of their production life.

[50] See I*nitiative for Heavily Indebted Poor Countries – Status of Implementation*, Table 1.

[51] Ibid.

[52] The projection period begins in 2002, because only the loan-by-loan update of the DSA that will be conducted at the completion point will determine the final debt numbers for the period after the decision point.

historic growth rates and standard deviations, which are used to calibrate the shocks, are computed on the basis of data for the period 1992-2000. The historic scenario shows non-stabilizing debt dynamics for the two debt stock indicators, and high debt service ratios (solid thin line in Figure 1). The other stress tests (broken line) show large vulnerabilities to one-standard deviation GDP and export growth shocks, and to a less severe but simultaneous shock to all variables (combined negative shock). These shocks raise the debt ratios to clearly unsustainable levels, yet one-standard deviation shocks occur relatively frequent so that they cannot be considered 'extreme'.

A possible explanation for the large vulnerability to shocks is that during the entire period 1992-2000, which was used to calibrate the shocks, Sierra Leone was engulfed in civil war. In this sense, the stress tests could reveal nothing more than that should the country slide back into war, its current debt will become once again unsustainable. To test this, a second alternative scenario has been computed, where all shocks have been calibrated based on historic data for the 1980s. The results, however, confirm those from the first set of stress tests: debt dynamics become unsustainable even if key variables grow at their historic rates from the 1980s (Table 1b). In sum, the vulnerability to shocks makes it necessary to consider Sierra Leone not at low, but at moderate risk of debt distress, even after the full delivery of HIPC relief. Maintaining low debt ratios is necessary to protect against the consequences of negative shocks, which have occurred frequently in the past. To achieve this, it will be necessary to continue to borrow on highly concessional terms and to seek grants whenever possible.

Box 1: Update of Sierra Leone's Debt Sustainability Analysis (DSA)

The update of Sierra Leone's debt sustainability analysis was done in two steps:

1. Sierra Leone's decision point DSA was updated to reflect end-2003 discount and exchange rates, and new borrowing and export assumptions:

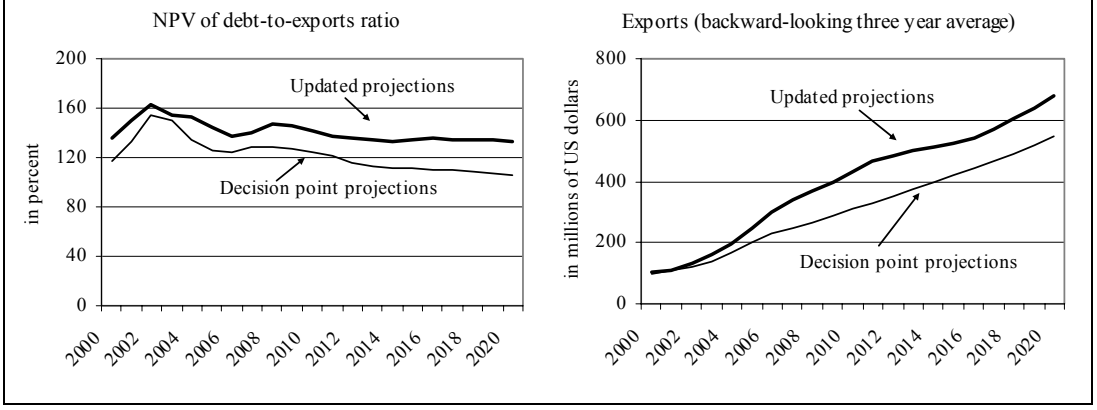
   a. The decline in the average discount rate by about 170 basis points and the depreciation in the US dollar by about 15 percent against major currencies since end-2000 (i.e., the base year for the decision point DSA) led to an increase in the NPV of debt by about 30 percent.

   b. New borrowing in the period 2001-04 is somewhat lower than expected, but medium-term borrowing projections have been revised upwards significantly. The decision point DSA had assumed a sharp decline in average annual new borrowing from about US$110 million in 2001-04 to US$45 million in the medium-term, whereas the updated DSA assumes that new borrowing over the medium term will approximately remain at 2001-04 levels of about US$90 million per annum, partly because absorptive capacity has increased recently (new borrowing is likely to reach US$130 million in 2004).

   c. Export performance in 2001-04 was largely as expected, but medium term projections have been revised upwards, reflecting, inter alia, the beginning of large-scale diamond mining (Koidu kimberlite mine). This partly offsets the upward revision in new borrowing.

   In sum, the updated NPV of debt-to-exports ratio is somewhat higher than projected at the decision point, with the NPV of debt-to-exports ratio leveling off now at about 135 percent (Figure 1).

Figure 1: Comparison of updated projections with decision point projections



Box 1: continued

2. In a second step, the methodology of the new DSA framework for low-income countries has been applied. Debt service and export projections are used as inputs into a template that computes baseline projections for the NPV of debt-to-exports ratio and then subjects the baseline to various stress tests. These baseline projections for the NPV of debt-to-export ratio differ from those computed in the first step under the HIPC methodology in two ways (Figure 2):

   a. The template uses a discount rate of 5 percent, which is about 60 basis points higher than the currency-specific discount rates used under HIPC. This lowers the NPV of debt by about five percent relative to the HIPC methodology. (Figure 2, broken line)

   b. The NPV of debt-to-exports ratio is computed relative to current exports, as opposed to the backward-looking three year average used under the HIPC methodology. In years when exports are growing rapidly, for example in 2000-06, the current-exports series is higher by 20 percent than the series for the backward-looking average. In contrast, in years when export growth is flat, i.e. in the period 2013-15 when the Koidu kimberlite mine reaches the end of its production life, the difference is small. (Figure 2, solid thin line)

Figure 2: NPV of debt-to-exports ratio under new DSA and HIPC methodologies

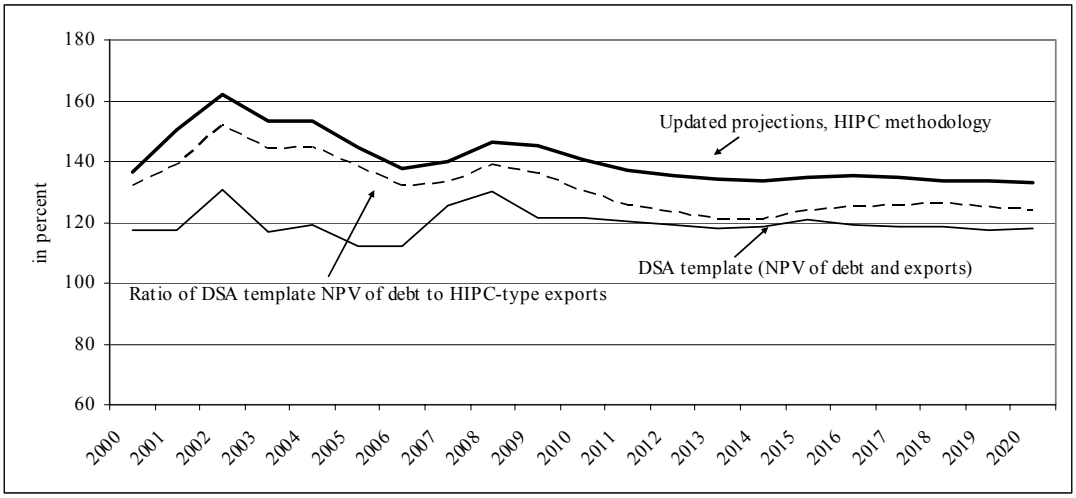


Table 1a. Country: External Debt Sustainability Framework, Baseline Scenario, 2000-2023 1/
(In percent of GDP, unless otherwise indicated)

	Actual				Historical Average 6/	Standard Deviation 6/	Estimate		Projections							
	# 1997	1998	1999	2000 7/			2001 8/	2002 9/	2003	2004	2005	2006	2003-08 Average	2011	2021	2009-23 Average
External debt (nominal) 1/	**134.6**	**175.4**	**182.8**	**37.5**			**34.4**	**37.5**	**41.2**	**49.7**	**53.3**	**55.2**		**57.2**	**45.3**	
o/w public and publicly guaranteed (PPG)	134.6	175.4	182.8	37.5			34.4	37.5	40.0	47.9	50.3	53.0		57.2	45.3	
Change in external debt	16.2	40.8	7.4	-145.3			-3.1	3.0	3.8	8.5	3.6	1.9		-0.9	-0.9	
Identified net debt-creating flows	13.0	38.1	13.1	23.5			15.7	-4.2	3.4	6.2	3.1	1.4		0.2	-0.3	
Non-interest current account deficit	**-0.4**	**0.4**	**8.6**	**12.9**	**4.6**	**7.1**	**16.0**	**4.3**	**7.2**	**10.8**	**7.2**	**4.2**		**2.1**	**1.3**	2.9
Deficit in balance of goods and services	1.2	2.8	15.7	21.3			19.6	19.5	21.3	23.2	17.2	12.7		7.3	3.2	
Exports	13.9	16.2	12.6	18.1			17.2	16.4	19.9	22.8	26.2	27.7		26.3	21.3	
Imports	15.2	19.0	28.3	39.4			36.9	35.9	41.2	46.1	43.4	40.4		33.6	24.5	
Net current transfers (negative = inflow)	-2.9	-5.5	-8.3	-9.2	-5.3	2.5	-5.5	-17.9	-16.5	-14.3	-11.6	-10.5		-6.3	-2.5	-5.1
Other current account flows (negative = net inflow)	1.3	3.0	1.2	0.8			2.0	2.7	2.3	1.9	1.6	1.9		1.2	0.6	
Net FDI (negative = inflow)	**-0.2**	**0.0**	**-0.2**	**-0.2**	**-0.1**	**0.7**	**-0.1**	**-0.2**	**-0.9**	**-2.4**	**-1.0**	**-0.1**		**-0.1**	**0.0**	-0.2
Endogenous debt dynamics 2/	**13.6**	**37.8**	**4.8**	**10.8**			**-0.3**	**-8.4**	**-2.9**	**-2.2**	**-3.2**	**-2.7**		**-1.9**	**-1.6**	
Contribution from nominal interest rate	0.8	2.2	2.5	2.2			1.4	0.5	0.4	0.7	1.0	0.8		0.6	0.4	
Contribution from real GDP growth	23.1	1.4	14.4	-7.3			-1.7	-8.8	-3.3	-2.9	-4.2	-3.5		-2.4	-2.0	
Contribution from price and exchange rate changes	-10.3	34.1	-12.2	15.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**3.2**	**2.7**	**-5.6**	**-168.9**			**-18.8**	**7.3**	**0.3**	**2.2**	**0.6**	**0.5**		**-1.1**	**-0.6**	
o/w exceptional financing	-1.4	0.0	0.0	0.0			-34.2	-2.6	-5.0	-4.0	-3.0	-2.8		-1.9	-0.6	
NPV of external debt 4/	...	...	...	21.3			20.3	21.4	24.5	29.1	32.4	33.4		31.9	24.7	
In percent of exports	...	...	...	117.5			117.7	130.8	123.0	127.2	123.7	120.5		121.2	115.9	
NPV of PPG external debt	**...**	**...**	**...**	**21.3**			**20.3**	**21.4**	**23.3**	**27.2**	**29.4**	**31.2**		**31.9**	**24.7**	
In percent of exports	...	...	...	117.5			117.7	130.8	116.8	119.1	112.1	112.6		121.2	115.9	
Debt service-to-exports ratio (in percent)	18.9	41.4	57.2	45.9			72.3	9.4	7.2	11.0	7.9	4.7		11.9	3.7	
PPG debt service-to-exports ratio (in percent)	18.9	41.4	57.2	45.9			72.3	9.4	7.2	10.4	7.5	2.8		8.7	3.7	
Total gross financing need (billions of U.S. dollars)	0.0	0.0	0.1	0.1			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	-16.5	-40.4	1.1	158.2			19.1	1.3	3.4	2.4	3.6	2.3		3.1	2.2	
Key macroeconomic assumptions																
Real GDP growth (in percent)	-17.6	-0.8	-8.1	3.8	-1.9	7.0	5.4	32.1	9.3	7.4	9.4	7.1	11.8	4.5	4.5	4.8
GDP deflator in US dollar terms (change in percent)	9.5	-20.2	7.5	-8.0	3.5	12.1	12.1	-5.4	-3.2	-1.8	1.6	1.7	0.8	2.5	2.0	2.2
Effective interest rate (percent) 5/	0.6	1.3	1.4	1.2	1.6	1.7	4.5	1.7	1.1	1.8	2.3	1.6	2.2	1.1	0.8	1.0
Growth of exports of G&S (US dollar terms, in percent)	-29.4	-8.0	-23.3	37.4	-4.3	22.2	12.3	18.6	28.8	20.8	27.5	15.1	20.5	4.2	6.8	5.4
Growth of imports of G&S (US dollar terms, in percent)	-61.4	-0.6	46.6	33.1	9.6	31.1	10.5	21.6	21.6	17.7	4.7	1.4	12.9	3.7	3.8	3.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	...	56.9	46.0	48.0	47.1	42.2	46.4	47.8	56.9	56.9	54.9
Memorandym item:																
Nominal GDP (billions of US dollars)	0.8	0.7	0.7	0.6			0.7	0.9	1.0	1.0	1.2	1.3		1.8	3.6	

Source: Staff simulations.

1/ Includes both public and private sector external debt. From 2000 onwards unconditional delivery of HIPC relief is assumed.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value. For public and public guaranteed debt, unconditional delivery of HIPC relief from 2000 onwards is assumed.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ The reduction in the debt stock in 2000 is due to the assumption of the unconditional delivery of HIPC relief. The hypothetical delivery of HIPC relief is not included in the exceptional financing item, and leads to a large unexplained residual.
8/ The large exceptional financing item in 2001 reflects a Paris Club rescheduling on Naples terms. The large debt service ratio is due to the repayment of post-emergency assistance to the Fund.
9/ In 2002, the BOP was compiled for the first time on the basis of the Balance of Payments Manual Fifth Edition, which led to a large upward revision in private transfers. Another major change ocurred in the compilation of national accounts data, where the use of new survey instruments has led to better measurement of previously unrecorded activity, which resulted in a large upward revision in GDP.

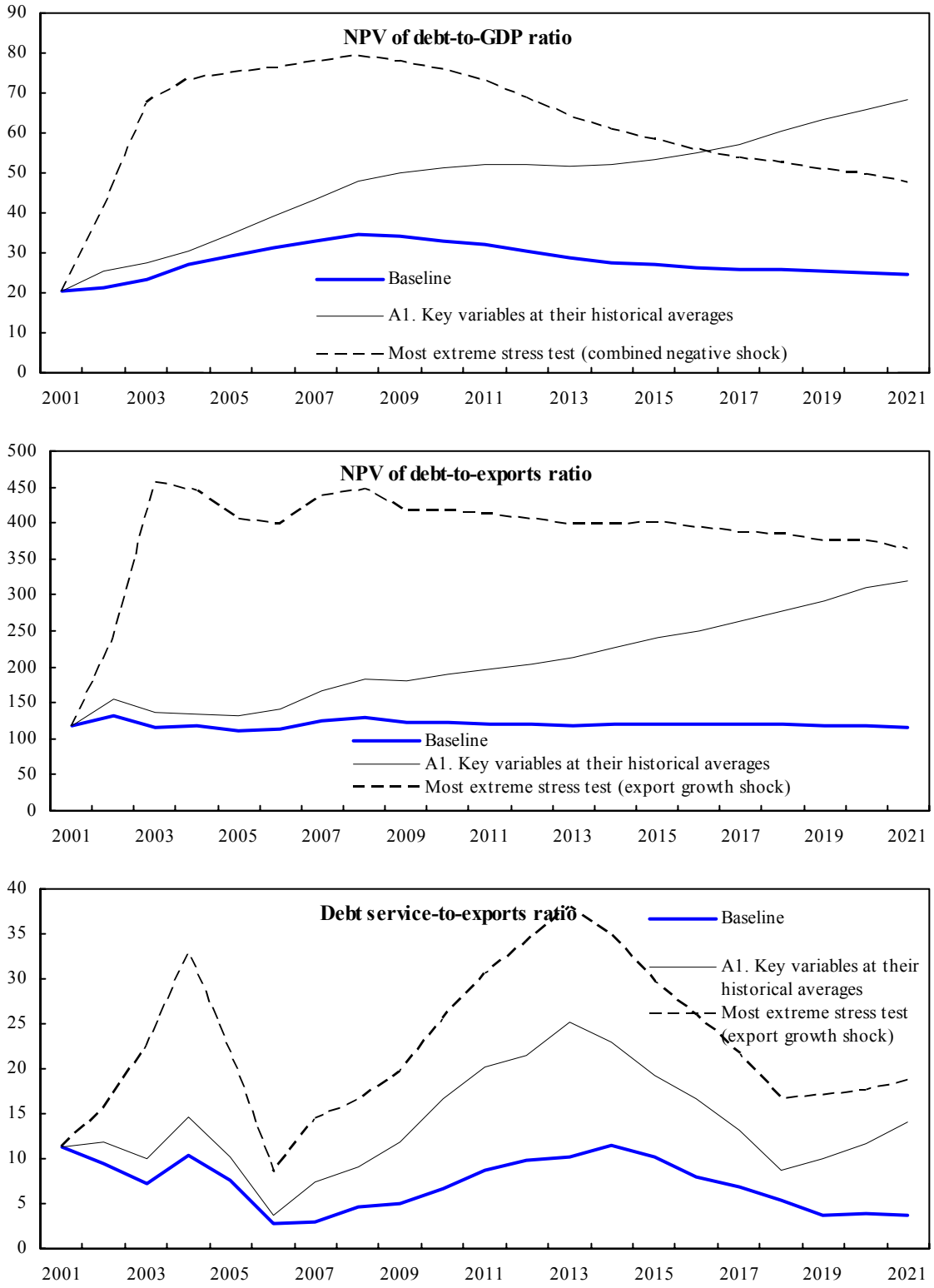
Table 1b. Country: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2003-23
(In percent)

	Estimate			Projections				
	2001	2002	2003	2004	2005	2006	**2011**	2021
NPV of debt-to-GDP ratio								
Baseline	20	21	23	27	29	31	**32**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages of the 1990s in 2002-21 1/	20	26	27	30	34	39	**52**	68
A2. Key variables at their historical averages of the 1980s in 2002-21 1/	20	26	28	31	35	41	**56**	76
A3. New public sector loans on less favorable terms in 2002-21 2/	20	24	27	33	37	39	**46**	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2002-03	20	31	40	47	51	54	**56**	43
B2. Export value growth at historical average minus one standard deviation in 2002-03 3/	20	24	32	36	38	39	**38**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2002-03	20	22	26	30	32	34	**35**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2002-03 4/	20	29	37	41	43	44	**42**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	20	41	68	73	75	76	**73**	48
B6. One-time 30 percent nominal depreciation relative to the baseline in 2002 5/	20	31	34	40	43	46	**47**	36
NPV of debt-to-exports ratio								
Baseline	118	131	117	119	112	113	**121**	116
A. Alternative Scenarios								
A1. Key variables at their historical averages of the 1990s in 2002-21 1/	118	156	138	133	132	141	**197**	321
A2. Key variables at their historical averages of the 1980s in 2002-21 1/	118	157	140	136	135	146	**213**	358
A3. New public sector loans on less favorable terms in 2002-21 2/	118	144	134	145	140	142	**173**	171
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2002-03	118	131	117	119	112	113	**121**	116
B2. Export value growth at historical average minus one standard deviation in 2002-03 3/	118	240	455	445	405	398	**413**	363
B3. US dollar GDP deflator at historical average minus one standard deviation in 2002-03	118	131	117	119	112	113	**121**	116
B4. Net non-debt creating flows at historical average minus one standard deviation in 2002-03 4/	118	176	188	181	162	158	**161**	135
B5. Combination of B1-B4 using one-half standard deviation shocks	118	261	466	441	392	379	**379**	307
B6. One-time 30 percent nominal depreciation relative to the baseline in 2002 5/	118	131	117	119	112	113	**121**	116
Debt service ratio								
Baseline 7/	11	9	7	10	8	3	**9**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages of the 1990s in 2002-21 1/	11	12	10	15	10	4	**20**	14
A2. Key variables at their historical averages of the 1980s in 2002-21 1/	11	12	10	15	11	4	**23**	17
A3. New public sector loans on less favorable terms in 2002-21 2/	11	10	9	12	9	5	**7**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2002-03	11	10	8	11	7	3	**10**	5
B2. Export value growth at historical average minus one standard deviation in 2002-03 3/	11	16	23	33	22	9	**31**	19
B3. US dollar GDP deflator at historical average minus one standard deviation in 2002-03	11	10	8	11	7	3	**10**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2002-03 4/	11	10	8	12	8	3	**12**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	11	14	18	26	18	8	**26**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2002 5/	11	10	8	11	7	3	**10**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	54	54	54	54	54	54	**54**	54

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly ass an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.
7/ For comparability, the debt service ratio for 2001 excludes the effect of the repayment of post-emergency assistance to the Fund.

Figure 1. Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2001-21



Source: Staff Projections and simulations.